

Business Proposal: AI-Powered Customer Support Chatbot

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Company:

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Executive Summary

We propose to develop and deploy an **AI-powered customer support chatbot** for our e-commerce platform to reduce support costs, improve response time, and increase customer satisfaction. The chatbot will handle 60% of routine inquiries, allowing our support team to focus on complex issues.

Total Investment:	\$85,000
Expected ROI:	220% in Year 1
Payback Period:	4.5 months

Problem Statement

Current Challenges:

- Support Team Overload: 12-person team handles 500+ inquiries daily → 24–48 hour delays
- High Operating Costs: \$480,000/year (salaries + tools + infrastructure)
- Poor CSAT: 35% report slow responses; NPS 42 (industry avg: 65)
- Scalability Issues: Adding staff costs \$55,000/year per person

Proposed Solution

Features:

- Multilingual Support (English, Spanish, French, Mandarin)
- NLU with Groq Llama 3.1 LLM
- Real-time order database integration
- Human escalation for complex issues
- 24/7 availability across time zones
- Machine learning improvements over time

Financial Projections

Metric	Year 1 Impact
Cost Savings (reduced team)	\$233,500
Revenue Gains (retention + NPS)	\$300,000

Total Benefit	\$533,500
ROI	528%
Payback Period	1.1 months

Implementation Timeline

Phase	Duration	Key Activities
Planning & Setup	2 weeks	Requirements, architecture, AWS setup
Development	8 weeks	Build backend, train model, integrate
Testing	3 weeks	QA, UAT, beta deployment (10% traffic)
Launch	2 weeks	Full rollout, team training, monitoring
Total	15 weeks	Go-live: mid-May 2026

Risk Assessment

- Poor AI accuracy → Mitigate: Start simple, human escalation
- Integration delays → Mitigate: API exists, add buffer time
- Rate limits → Mitigate: Queuing system for spikes
- Data security → Mitigate: Encryption, GDPR compliance

Conclusion

The AI chatbot is a **low-risk, high-return investment** with a 4.5-month payback period and 220% Year 1 ROI. Recommend immediate approval. Target launch: May 2026.